

- (endocrinology), Morehouse and Miller (physiology of exercise), and Bacq and Alexander (radiobiology) are the leading textbooks.
8. About a third of the institutions have animal room, stock room, and instrument workshop facilities exclusively for physiology use.
 9. The commonest apparatus and the greatest frequency of use were reported for the kymograph, compound microscope, oscilloscope, centrifuge, multi-channel recording apparatus (polygraph), analytical balance, and spirometer.
 10. The most popular laboratory animals and the greatest frequency of use were recorded for the "frog," "turtle," "rat," and "rabbit," with percentages of 97, 90, 90, and 64, respectively.

REFERENCE

1. Henzlik, Raymond E. "Mechanics of Organization in Introductory Zoology Courses in State Universities and Provisions for the 'Gifted' Students." *Science Education*, 47:462-468, December, 1963.

DIALOGUE BETWEEN THE EARTH AND THE MOON

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EARTH. Dear Moon, I know that you can talk and answer questions for you are a person, as I have heard many times from the poets. And besides, our children say that you really have a mouth, nose and eyes as each of them has; and that they see them with their own eyes which at their age it is reasonable to suppose are very sharp. As for me, I don't doubt that you know that I too am no more and no less a person, especially since when I was younger I gave birth to many sons.² So you shan't be surprised at hearing me speak. Well then, my fine Moon, although I've been near you for so many centuries that I don't remember the number, I've never said a word to you until now, because my affairs kept me so busy that there was no time left for chatting. But today, since my occupations are very much reduced (actually, I can say that they go independently), I don't know what to do

with myself, and I'm bursting with boredom. Therefore, I'm planning from now on to converse often with you, and to worry a lot about your affairs, if it doesn't bother you.

MOON. Don't hesitate on that account. Would that I were as sure that luck would save me from every other annoyance as I am that you'll give me none! If you wish to converse with me, converse with me as much as you like. Although I'm a friend of silence, as I think you know, I shall listen to you and shall be glad to answer your questions, to be of service to you.

EARTH. Do you hear this very agreeable sound which the celestial bodies make with their motions?

MOON. To tell you the truth, I don't hear anything.

EARTH. I don't hear anything either except for that din of the wind which blows from my poles to the equator and from the equator to the poles, and it is not at all musical. But Pythagoras says that the celestial spheres make a certain sound so sweet that it is a marvel, and that you too have a part in it. You are the eighth chord of this cosmic lyre. But he says that I'm absorbed by the sound itself and therefore don't hear it.

MOON. No doubt I'm deafened too. As

¹ This translation of Leopardi's *Dialogo della Terra e della Luna* is Rufus Suter (deceased, February 29, 1968). The dialogue originally appeared in 1827. The text used for the translation is in Leopardi's *Operette morali*, Rizzoli-Editore, Milan (Biblioteca Universale Rizzoli, 327-329), first ed., 1951: pp. 55-62; 292. The following notes (unless otherwise indicated) are translated from this text.

² The earth was regarded as the mother of the Titans.

I said, I don't hear it. And I don't know anything about being a chord.

EARTH. Then let's change the subject. Tell me: are you really inhabited as a thousand philosophers, ancient and modern, affirm and swear—from Orpheus to De la Lande?³ In spite of all my efforts to stretch these horns of mine, which men call mountains and peaks and with the points of which I make a practice of looking at you after the habit of a large snail, I have never succeeded in discovering a single inhabitant on you even though I hear that a certain David Fabricius,⁴ who could see better than Lynceus,⁵ once discovered some persons on you who were spreading a wash out to dry in the sun.

MOON. About your horns I don't know what to say. The fact is, I am inhabited.

EARTH. Of what color are those men of yours?

MOON. What men?

EARTH. Those men who are on you. Don't you say that you are inhabited?

MOON. Yes. And so?

EARTH. And so your inhabitants certainly won't all be animals.

MOON. They're neither animals nor men. I don't know what breed of creature either is. And indeed of the several things which you have been hinting to me apropos (I suppose) of men, I haven't understood a jot or a tittle.

EARTH. But what sort of a population are they?

³ [Translator's note: Though, of course, this dialogue is purely a fanciful work of the imagination, Leopardi, to give it the semblance of fact in a humorous way, cites actual historical personalities such as] Joseph Jérôme De la Lande (1732-1807), French astronomer who sustained the habitability of the planets.

⁴ David Fabricius (1564-1617), Dutch astronomer who not only sustained the habitability of the worlds, but also insisted that he had seen with his own eyes the inhabitants of the moon.

⁵ A legendary Greek hero whose vision was so sharp that he could see through earth and stones. Ho took part in the expedition of the Argonauts. [Translator's note: Possibly the famous Italian Accademia dei Lincei to which Galileo belonged was named after this character, although "lincei" also means lynxes which likewise are known for their sharp sight.]

MOON. They are very numerous and they are of very many different kinds. You're not acquainted with any of them, as I'm not acquainted with any of yours.

EARTH. That seems so strange to me that if I hadn't heard it from you personally I shouldn't have believed it for any reason in the world. Were you ever conquered by any of yours?

MOON. No. Not as far as I know. How? Why?

EARTH. For ambition. For greed. With political skills. With arms.

MOON. I don't know what "arms," "ambition," "political skills" mean. In a word I don't know what you're talking about.

EARTH. But surely if you aren't acquainted with arms, you are acquainted at any rate with war, for a short time ago a physicist down here with certain telescopes (which are instruments made to see a very great distance) discovered a fine fortress there on you with upright bastions.⁶ This is a sign that your folk are accustomed at least to sieges and to wall-battles.

MOON. Pardon, Lady Earth, if I reply to you a bit more freely than perhaps is fitting for your subject and maid-servant, as I am. But really you seem to me worse than conceited in thinking that everything in every part of the world conforms to your things as if nature had no other purpose than to copy you everywhere, point for point. I speak of being inhabited and

⁶ Professor Gruithuisen of Munich, who was well known for his selenological studies, described a colossal building similar to a fortress, near the moon's equator. It had upright bastions arranged like the veins of an alder leaf. Also he discovered many excellently constructed highways on the moon, unmistakable signs of seasonal cultivation, and other traces of intelligent life on that planet. Gruithuisen's findings were reported in the astronomical annals of Johann Bode (1747-1826), director of the Berlin Observatory; in reports by physicists of Bonn; and in other publications. His idea about signs of seasonal cultivation on the moon he shared with Johann Schröter (1745-1816) of Erfurt, the founder of selenography. The telescopes which Gruithuisen used for his discoveries were made by Josef Fraunhofer (1787-1826) [Translator's note: after whom the Fraunhofer lines of the spectrum are named].

you conclude from this that my inhabitants must be men. I inform you that they aren't, and you, while acknowledging that they are other creatures, don't doubt that they have the same qualities and participate in the same events as your population. And you allege to me the telescopes of I-don't-know-what physicist. But if those telescopes of yours don't see any better in other matters I'll believe that they have the good vision of your children who see on me eyes, a mouth, a nose I don't know where.

EARTH. Then it isn't even true that your provinces are furnished with broad and trim highways and that you are cultivated—things which are clearly seen from Germany when one picks up a telescope.

MOON. Whether I'm cultivated I'm not aware; and I don't see my highways.

EARTH. Dear Moon, you must know that I'm simple and innocent and it's no wonder that men easily deceive me. But I can tell you that if yours don't care about conquering you, you were not for this reason always safe from danger. For at various times many persons down here have had the idea to conquer you. To this effect they made many preparations. But when they had ascended into very high places and had raised themselves on the tips of their toes and had stretched up their arms they couldn't reach you. Besides this, already for not a few years I've seen a minutely scrupulous spying on all your sites, a drawing of maps of your regions, a measuring of the heights of those mountains of yours of which we know even the names.⁷ For the sake of the good will which I bear towards you it has seemed to me well to inform you about these things in order that you won't fail to take precautions for any eventuality. Now, to change the subject: Are you bothered by the dogs that bark at you? What do you think of those who show you to others in

a pond? Are you masculine or feminine (because in ancient times there was a variety of opinion on this subject)? Is it true or not that the Arcadians came into the world before you? That your women (or whatever I ought to call them) are egg-bearing? And that one of their eggs fell down here I don't know when? That you are pierced like the Lord's Prayer beads in a Rosary, as a modern physicist believes? That you are made of green cheese as some Englishmen suppose? That one day or night (whichever it was) Mohammed split you through the middle like a watermelon and that a good chunk of your body slipped up one of his sleeves? Do you enjoy standing on the tips of minarets? What is your opinion of the Festival of Bairam?

MOON. Just go ahead, for while you continue thus I have no reason to answer you or to break my customary silence. If you love to entertain yourself with prattle and don't find other topics than these, rather than turning to me who can't understand you it will be better for you to have men manufacture another planet for you, to whirl around you,⁸ which will be composed and inhabited after your fashion. You don't know how to talk about anything except men, dogs and similar things, of which I have as much information as I do of that great, great sun around which I hear that you whirl.

EARTH. It's true that the more I resolve not to touch upon my own affairs when I converse with you, the less I succeed. But henceforth I'll be more careful. Tell me: Is it you who amuse yourself by pulling the water of my sea up and then letting it fall down again?

MOON. Could be. But granted that I exert that or some other effect on you, I'm not conscious of so doing; as you similarly, as far as I can tell, are unaware

⁷ [Translator's note: Earth's warning, of course, was prophetic. Not only is the map-making still in process, but the plans to invade the moon are moving ahead very rapidly.]

⁸ [Translator's note: Earth has decided, somewhat belatedly, to follow her maid-servant's advice. Witness the scores of man-made moons now whirling around her.]

of many effects which you exert here. And they must be as much greater than mine as you are greater than me in size and strength.

EARTH. About these effects I really know nothing except that now and then I take away the sun's light from you and your light from me; and again that I make a great light for you during your nights, which in part I sometimes see. But I was forgetting one thing which is more important than all the rest. I should like to know whether really, according to what Ariosto⁹ writes, everything which each man keeps on losing, such as youth, good looks, health, the efforts and money which are put in worthy studies for the sake of being honored by others, for directing children towards good habits, for making and promoting useful institutions—whether all this goes up there and collects on you, so that all things human are found on you with the exception of craziness, which does not part from men. In case that this is true, I realize that you must be so full that you have no room left, especially since in recent times men have lost very many things (*e.g.*, patriotism, virtue, magnanimity, rectitude) and not just in part, and one or another of them as previously, but everything and everybody. Certainly, if these lost things are not up there on you, I don't believe that they can be anywhere else. Therefore, I should like to make a pact with you by which you would immediately hand back to me all these things, one by one. I think that you yourself might be glad to be disencumbered especially of the good sense which I understand occupies a very large space up there on you. And I'll have men pay a handsome sum of money to you each year.

MOON. You return to men. And although craziness, as you affirm, doesn't part from your boundaries, you want to drive me crazy in every way and to take

away my power of judgment by seeking theirs. I don't know where their power of judgment is, nor whether it goes or halts anywhere in the world. I know well that it isn't here, as the other things you inquire about aren't here.

EARTH. At least you will be able to tell me whether up there on you the vices, misdeeds, hard luck, griefs, old age, in short the evils, are in use. Do you understand these nouns?

MOON. O those nouns! Yes, I understand them. Not only the nouns, but the things signified I'm marvelously well acquainted with because I'm all full of them instead of with those other things which you thought were here.

EARTH. Which prevail in your population: merits or defects?

MOON. The defects, by far.

EARTH. Of which do you have the greater abundance: goods or ills?

MOON. The ills, incomparably.

EARTH. In general, are your inhabitants happy or unhappy?

MOON. So unhappy that I shouldn't exchange places with the luckiest of them.

EARTH. It's the same here. I marvel that you can be so different from me in other respects and so like me in this.

MOON. Also in shape and in my whirling and in being illuminated by the sun I conform to you. The likeness you speak of is no more to be marveled at than these conformities. For evil is common to all the planets of the universe, or at least of this solar system, just like the sphericity and the other conditions of which I've spoken—no more and no less. If you could raise your voice high enough to be heard by Uranus or Saturn or any other planets of our solar system, and if you asked whether on them unhappiness had a place, and whether the goods prevailed over or yielded to the evils, each would reply to you as I have. I say this because I've asked the same questions of Venus and Mercury, to which planets I find myself from time to

⁹ See Ariosto's *Orlando furioso*, canto 34. Ariosto imagined that everything which men lose, including their good sense, collects on the moon.

time nearer than you do. Also I have made the same inquiry of some comets who pass my way. All have answered as I have done. I believe that even the sun himself and every star would answer in the same way.

EARTH. Despite all that you say I have hopes. Today in particular men promise me much happiness in the future.

MOON. Hope as much as you like. And I promise you that you can hope forever.

EARTH. You know what? These men and these animals are starting to make a noise, for at the spot from which I'm con-

versing with you it's night, as you see, or rather as you don't see. They were all sleeping and at the din which we are making with our talking they are awakening in great fear.

MOON. But here, as you see, it's day.

EARTH. Now I don't want to be the cause of terrifying my folk and of interrupting their sleep, which is the greatest good they have. However, we'll talk to each other again some later time. So good-bye. Good day.

MOON. Good-bye. Good night.

GIACOMO LEOPARDI: DIALOGUE BETWEEN FREDERICK RUYSCH AND HIS MUMMIES¹

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CHORUS OF CORPSES IN FREDERICK RUYSCH'S LABORATORY²:

*"In you, alone eternal in the world, to whom
Turns every created thing,
Rests, O Death,
Our shorn being,
Not happy, but safe
From old griefs. Deep night
Dims the sober thought
Of our confused mind;
Our barren spirit feels itself
Too weak to hope, to wish.
Thus, free of sorrow and of fear,*

*It consumes the slow and empty
Ages without ennui.
We lived; and as the confused memory
Of a hideous ghost
Or of a feverish nightmare
Wanders in the suckling infant's soul,
So lingers the memory
Of our life; but the memory
Is remote from fear. What were we?
What was that bitter point
Which had the name of life?
A secret and stupefying thing
Is life now to our thoughts, and such as
Unknown death seems
To the thought of the living. As from death
Our live being flees, so flees
From the vital flame
Our shorn being,
Not happy, but safe.
However, to be blessed
Is denied by Fate to mortals and to the
dead."*

¹ This translation of Leopardi's *Dialogo di Federico Ruysch e delle sue mummie* is by Rufus Suter (deceased, February 29, 1968). The dialogue appeared originally in 1827. The text used for this translation is in Leopardi's *Operette morali*, Rizzoli-Editore, Milan (Biblioteca Universale Rizzoli, 327-329), first ed., 1951, pp. 130-136. The following notes, unless otherwise indicated, are taken from the text.

² "In regard to these mummies, which in scientific language would be called anatomical preparations, see Fontenelle, *Éloge de Mons. Ruysch*" (Leopardi). Ruysch was a physician of Amsterdam (1638-1731), who succeeded in preserving cadavers from corruption.

RUYSCH (OUTSIDE HIS LABORATORY
LOOKING THROUGH THE CRACKS OF THE
DOOR). The deuce! Who has taught music